

1D CNN — Architecture Variant Comparison

Generated 2026-05-14 22:01:07

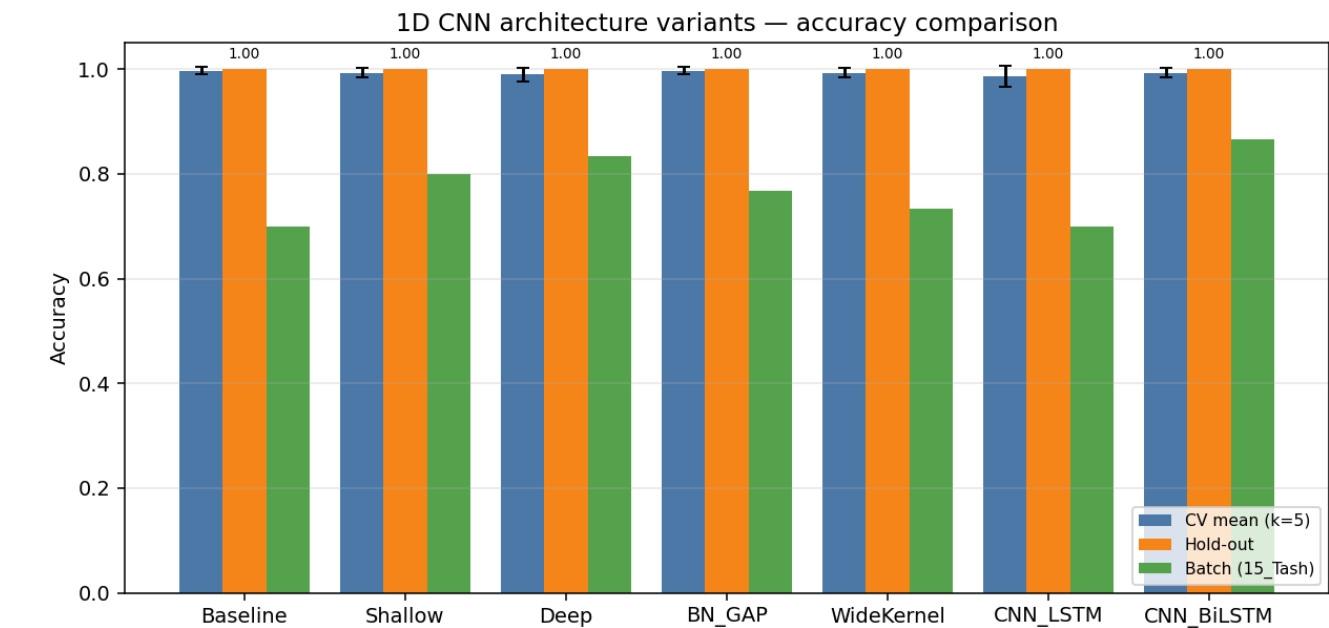
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Training path 11	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Total trials	325 (train pool 292, hold-out 33)
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · no transforms enabled
Epochs · batch size	60 · 16

Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,190	0.9966	0.0068	1.0000	0.0069	0.7000	21/30
Shallow	72,454	0.9932	0.0084	1.0000	0.0001	0.8000	24/30
Deep	203,878	0.9898	0.0136	1.0000	0.0000	0.8333	25/30
BN_GAP	57,190	0.9966	0.0068	1.0000	0.0000	0.7667	23/30
WideKernel	131,622	0.9932	0.0084	1.0000	0.0035	0.7333	22/30
CNN_LSTM	68,390	0.9864	0.0198	1.0000	0.0000	0.7000	21/30
CNN_BiLSTM	105,510	0.9932	0.0084	1.0000	0.0000	0.8667	26/30

Best on hold-out: **Baseline** · Best on CV: **Baseline**



Training curves (final retrain on full training pool)

Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,190
CV mean ± std	0.9966 ± 0.0068
Hold-out test acc / loss	1.0000 · 0.0069
Batch-test acc	0.7000 (21/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	1.0000	1.0000	1.0000
2	1398	59	0.9831	0.9831	0.9831
3	1404	58	1.0000	1.0000	1.0000
4	1404	58	1.0000	1.0000	1.0000
5	1404	58	1.0000	1.0000	0.8448

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	0	5	0.000
Drum_Roll	1	5	0.200
Overall	21	30	0.700

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,454
CV mean ± std	0.9932 ± 0.0084
Hold-out test acc / loss	1.0000 · 0.0001
Batch-test acc	0.8000 (24/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	1.0000	1.0000	0.9831
2	1398	59	0.9831	0.9831	0.8814
3	1404	58	1.0000	1.0000	0.9310
4	1404	58	0.9828	0.9828	0.9828
5	1404	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	3	5	0.600
Drum_Roll	1	5	0.200
Overall	24	30	0.800

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	203,878
CV mean ± std	0.9898 ± 0.0136
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.8333 (25/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	1.0000	1.0000	0.9831
2	1398	59	0.9661	0.9831	0.9831
3	1404	58	1.0000	1.0000	1.0000
4	1404	58	0.9828	0.9828	0.9828
5	1404	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	1	5	0.200
Overall	25	30	0.833

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
CV mean ± std	0.9966 ± 0.0068
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7667 (23/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	1.0000	1.0000	0.6610
2	1398	59	0.9831	0.9831	0.9661
3	1404	58	1.0000	1.0000	1.0000
4	1404	58	1.0000	1.0000	1.0000
5	1404	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	1	5	0.200
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	2	5	0.400
Overall	23	30	0.767

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,622
CV mean ± std	0.9932 ± 0.0084
Hold-out test acc / loss	1.0000 · 0.0035
Batch-test acc	0.7333 (22/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	1.0000	1.0000	1.0000
2	1398	59	0.9831	0.9831	0.9831
3	1404	58	1.0000	1.0000	1.0000
4	1404	58	0.9828	0.9828	0.9828
5	1404	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	4	5	0.800
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	2	5	0.400
Drum_Roll	1	5	0.200
Overall	22	30	0.733

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,390
CV mean ± std	0.9864 ± 0.0198
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.7000 (21/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	0.9492	0.9661	0.9661
2	1398	59	0.9831	0.9831	0.9831
3	1404	58	1.0000	1.0000	1.0000
4	1404	58	1.0000	1.0000	0.9655
5	1404	58	1.0000	1.0000	1.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	2	5	0.400
Double_cmere	1	5	0.200
Drum_Roll	3	5	0.600
Overall	21	30	0.700

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,510
CV mean ± std	0.9932 ± 0.0084
Hold-out test acc / loss	1.0000 · 0.0000
Batch-test acc	0.8667 (26/30)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1398	59	1.0000	1.0000	1.0000
2	1398	59	0.9831	0.9831	0.9661
3	1404	58	1.0000	1.0000	0.9828
4	1404	58	0.9828	0.9828	0.9828
5	1404	58	1.0000	1.0000	0.9828

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	33
weighted avg	1.000	1.000	1.000	33

Batch test per class (15_Tash)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	4	5	0.800
Double_cmere	4	5	0.800
Drum_Roll	3	5	0.600
Overall	26	30	0.867